Membership Inference Attack Against Time-Series Prediction Models

Qingwen Li, Xiao Han, Ruiyan Wang, Junjie Wu, and Lanjuan Liu

***Abstract*—Recent advances in machine learning (ML) have raised growing privacy concerns. One notable concern of ML models is the membership leakage risk, which is defined as the extent to which an adversary can accurately determine whether a record was part of a model's training set through membership inference attacks (MIAs). While extensively studied in non-time-series models, this risk remains largely unexplored for time-series prediction models. To address this gap, this work aims to provide a precise evaluation of membership leakage risks for time-series prediction models. We first conduct an empirical analysis to identify distinct temporal discrepancies within sequential outputs between member and non-member samples in time-series models. Beyond global temporal discrepancies, our findings show that member records exhibit notable local discrepancies, including smaller fluctuations in their output sequences when actual values remain constant and more rapid adjustments when actual values change, compared with non-member records. Building on these insights, we propose TSP-MIA, a framework that leverages a patch-based attack model to capture both global and local discrepancies for more effective MIAs. Extensive evaluations demonstrate the effectiveness and robustness of TSP-MIA, revealing substantial membership leakage risks for time-series prediction models. We further assess the validity of existing defense mechanisms against TSP-MIA.**

***Index Terms*—Membership inference attack, time-series prediction models, privacy risk evaluation.**

I. Introduction

RECENT advances in machine learning (ML) models have been driven by the availability of large-scale data. In high-stakes domains such as healthcare and finance, vast amounts of personal data are utilized for model training, raising significant privacy concerns. One notable concern is the membership leakage risk, which is defined as the extent to which an adversary can accurately determine whether a record was part of a model's training set through membership inference attacks (MIAs) [1], [2], [3], [4], [5]. Typically, membership leakage risk is quantified by the attack performance of MIAs, and a higher attack performance indicates a greater membership leakage risk faced by the model. Given data protection regulations such as HIPAA and GDPR [6], membership leakage risk evaluation of ML models is essential before deploying them in real-world applications to ensure regulatory compliance and mitigate potential privacy breaches.

MIAs, which serve as a basis for membership leakage risk evaluation, have been extensively studied across various ML models, such as classification [1], [4], regression [7], and generative models [8], [9], [10]. These studies analyze and exploit discrepancies in model outputs between member (i.e., training data) and non-member records to successfully perform MIAs. Specifically, ML models often memorize characteristics of their training data, leading to outputs with higher confidence scores [1] or greater output consistency of augmented inputs [8] for member records than non-member records. Consequently, MIAs can be more effective against a given ML model when adversaries identify and utilize suitable features to differentiate between member and non-member records.

Most existing research has focused on MIAs against non-time-series models, such as image classifiers [1], [3], GANs [9], [10], and diffusion models [11], [12], [13]. In contrast, this study investigates membership leakage risks for time-series prediction models under MIAs. Such evaluation is especially important given the widespread use of these models in high-stakes domains such as traffic forecasting [14], healthcare [15], and default prediction [16]. Moreover, adversaries can exploit these models to generate a sequence of temporally correlated predictions from a record's sequential input, as illustrated in Fig. 1. This sequential output provides richer information for MIAs, potentially leading to higher membership leakage risks for time-series models compared to non-time-series models [17]. Therefore, studying MIAs against time-series prediction models is essential for understanding and mitigating their membership leakage risks.

A recent study introduced an MIA framework for time-series models, leveraging temporal modeling techniques to analyze the models' sequential outputs [17]. This approach exploits general temporal patterns (i.e., seasonality and trends) embedded in the

Received 18 May 2025; revised 14 October 2025; accepted 19 November 2025. Date of publication 26 November 2025; date of current version 12 March 2026. The work of Xiao Han was supported in part by the National Key Research and Development Program of China under Grant 2023YFC3304700 and in part by the National Natural Science Foundation of China under Grant 72071125 and Grant 72031001. The work of Prof. Junjie Wu's was partially supported by the National Key R&D Program of China under Grant 2023YFC3304700, in part by the National Natural Science Foundation of China under Grant 72031001 and Grant 72242101, in part by the Outstanding Young Scientist Program of Beijing Universities under Grant JWZQ20240201002, and in part by the Artificial Intelligence Technology R&D Center for Exploration and Development of China National Petroleum Corporation under Grant 2024-KFKT-22. *(Corresponding author: Xiao Han.)*

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Digital Object Identifier 10.1109/TDSC.2025.3637371

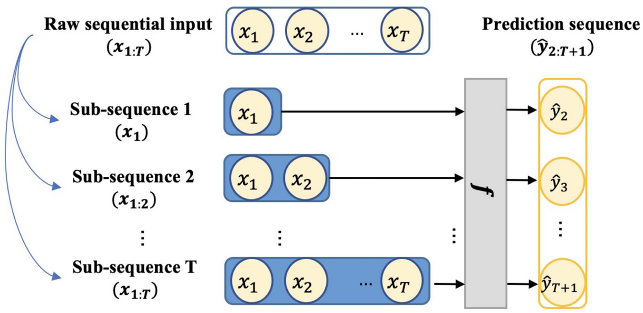


Fig. 1. An example of malicious exploitation of a time-series model for MIAs: Malicious adversaries may extract multiple subsequences from a sequence record and generate a series of temporally correlated outputs to create informative features for training an MIA classifier.

output sequences to distinguish members from non-members. The study suggests that the output sequences of member records typically align more closely with their actual values compared to those of non-member records. While this finding is insightful, trends and seasonality do not always manifest, particularly in shorter sequences, and often require long-term observations to be reliably detected [18]. This raises several important questions about MIAs in time-series models. For example, are there other universally present distinctions in sequential outputs that can differentiate member and non-member records? Additionally, how can we accurately model these distinguishing patterns to assess membership leakage risks fairly and reliably?

To address these questions, we begin with an empirical analysis to identify key differences in the sequential outputs of member and non-member records in time-series models, drawing insights from MIAs in non-time-series contexts. Our findings indicate that when the actual sequential values of a record change, the predicted output sequences of member records adjust more swiftly to these changes in the subsequent time steps, whereas those of non-members tend to lag behind or deviate from these changes. In contrast, when the actual values remain constant across steps, the predictions of member records tend to be more stable with lower variance than those of non-members. Previous studies have highlighted that model memorization and overfitting are key factors contributing to effective MIAs against non-time-series models [1], [19]. Our work extends this understanding to the temporal context, uncovering critical discrepancies that subsequently amplify the membership leakage risks associated with time-series prediction models.

Building on the above analysis, we aim to design a robust and effective MIA method that enables more accurate evaluation of membership leakage risk for time-series prediction models, leveraging the uncovered discrepancies. General temporal modeling architectures, such as RNN and LSTM, offer a straightforward approach to capturing patterns in sequential data. These methods are effective at learning global temporal dependencies (such as trends and seasonality) across the entire sequence and are well-suited for general time-series classification tasks [20], [21]. However, they may overlook critical local patterns, such as the observed rapid or gradual changes in a member or non-member record that occur over a short period before and after a key transition point in the sequence. Consequently, attack models constructed with RNN or LSTM may yield suboptimal MIA performance and thereby underestimate the membership leakage risks for time-series prediction models.

In this work, we propose TSP-MIA, a novel MIA framework that leverages emerging patch-based techniques [22], [23], [24] to capture both local and global temporal patterns in the sequential outputs of time-series models for effective MIAs. Specifically, TSP-MIA achieves this by ensembling multiple patch-based membership classifiers, each operating at a different temporal resolution: For a given patch size, a patch-based membership classifier first segments the sequential outputs into smaller patches (i.e., sub-sequences), which are then passed through a linear embedding layer to learn patch embeddings. These patch-based embeddings aim to preserve key local temporal characteristics over short time intervals, such as the more rapid adjustment observed in member records and the slower changes in non-member records when the actual values transit. To model global dependencies, the classifier incorporates a self-attention layer that models inter-patch temporal correlations. The resulting representation is finally passed to a binary classification head to predict membership status. Additionally, to capture discrepancies across multiple temporal resolutions, TSP-MIA ensembles the outputs of multiple patch-based classifiers with different patch sizes. The final predicted status is obtained by aggregating their outputs, leading to more effective MIAs and thus a more precise evaluation of membership leakage risks for time-series prediction models. In sum, this work makes the following major contributions:

- It highlights key temporal discrepancies in the sequential outputs between member and non-member records, enhancing the understanding of why and how MIAs succeed in time-series prediction models.
- It introduces TSP-MIA, an innovative MIA framework that integrates patch-based attack models to effectively leverage the identified temporal discrepancies, improving MIA performance and enabling a more accurate evaluation of membership leakage risks faced by time-series prediction models.

We conduct extensive experiments to evaluate TSP-MIA on two real-world datasets across a range of time-series prediction models with different architectures. The results show that TSP-MIA surpasses previous state-of-the-art baselines, uncovering significant membership leakage risks in time-series models that were previously underestimated. We further provide comprehensive analyses on different settings of model components and the adversary's knowledge of the target model, as well as the effectiveness of defenses against TSP-MIA, thereby offering deeper insights into the evaluation of membership leakage risks.

II. Background

A. Membership Inference Attack

Membership inference attacks (MIAs), first introduced by Shokri et al. [1], are among the most prevalent privacy attacks against machine learning (ML) models [25], [26], [27]. In a typical MIA scenario, an adversary attempts to infer whether a particular record was used to train a target model. Formally, given a record r and a trained target model f, an MIA can be expressed as:

$$\mathcal{A} : r, f \rightarrow \{0, 1\} \tag{1}$$

where $\mathcal{A}$ is the attack model, which is typically formulated as a binary classifier and takes different formats according to different task scenarios. If the record is part of the training dataset, the attack model outputs 1 (member); otherwise, it outputs 0 (non-member). Most prior studies assume a black-box attack setting, where the adversary only has access to the target model's output for a given record. To conduct MIAs in this setting, the adversary typically leverages output discrepancies between member and non-member records, generally arising from overfitting and model memorization [19], [28]. For instance, trained ML classifiers often assign higher confidence scores to member records compared to non-members [1], [19], [29]. Adversaries leverage these differences in confidence to construct membership features and train an attack model capable of predicting the membership status of a queried record. Intuitively, the greater the output discrepancy between members and non-members, the more effective the MIA, thereby indicating a greater membership leakage risk for the target model [30], [31].

MIA methods are broadly categorized into two types according to the construction of the attack model, i.e., classifier-based MIAs and metric-based MIAs [32]. Classifier-based MIAs involve training a binary classifier to distinguish the target model's behavior on members and non-members. Since the ground-truth membership labels of the target model are typically inaccessible, many prior works adopt the "shadow training" paradigm proposed by Shokri et al. [1]. Specifically, the adversary first constructs multiple shadow models to emulate the behavior of the target model. By collecting the outputs of these shadow models on both member (training) and non-member (test) records, paired with their corresponding binary membership labels, the adversary constructs a labeled dataset. This dataset is then used to train an attack model, typically a binary classifier, that learns to distinguish member from non-member records based on outputs. Once trained, the attack model is applied to the target model: by querying it with candidate records and analyzing the resulting outputs, the attack model predicts the likelihood that each record was included in the target model's training set. On the other hand, metric-based MIAs [3], [19], [28] infer membership status using a simpler, less computationally intensive approach. These approaches directly calculate membership metrics based on the target model's outputs and compare them against a predefined threshold to classify records as members or non-members. In this study, we exploit temporal discrepancies between the sequential outputs of members and non-members to enhance MIAs against time-series prediction models. Since classifier-based approaches enable the capture of these discrepancies more effectively by incorporating advanced modeling techniques, we adopt a classifier-based design.

Similar to our work, a recent study by Koren et al. [17] made the first attempt for MIAs against time-series forecasting models. They attempt to distinguish members from non-members based on temporal patterns such as seasonality and trends in the sequential outputs. However, these patterns do not always appear in a sequence and often require long-term observations for reliable detection. As a result, their approach may lead to limited MIA performance and underestimate membership leakage risks. In contrast, our study begins with an empirical analysis to identify key temporal discrepancies that are universally present in the sequential outputs of members and non-members in time-series models. We then propose TSP-MIA, a specialized MIA framework designed to accurately capture the identified discrepancies, thereby enabling a more precise assessment of membership leakage risks in time-series prediction models.

B. Time-Series Predictive Analytics

Time-series predictive analytics aims to exploit temporal patterns of historical records to predict future events or trends, serving as a powerful tool for various downstream applications such as traffic flow forecasting [14], intelligent healthcare [15], and default prediction [16]. Formally, given a sequence of historical observations $X = \{x_1, \ldots, x_T\}$ as the input, a standard time-series prediction task can be formulated as:

$$f_\theta : \{\mathrm{x}_1, \ldots, \mathrm{x}_T\} \rightarrow \hat{\mathrm{y}} \tag{2}$$

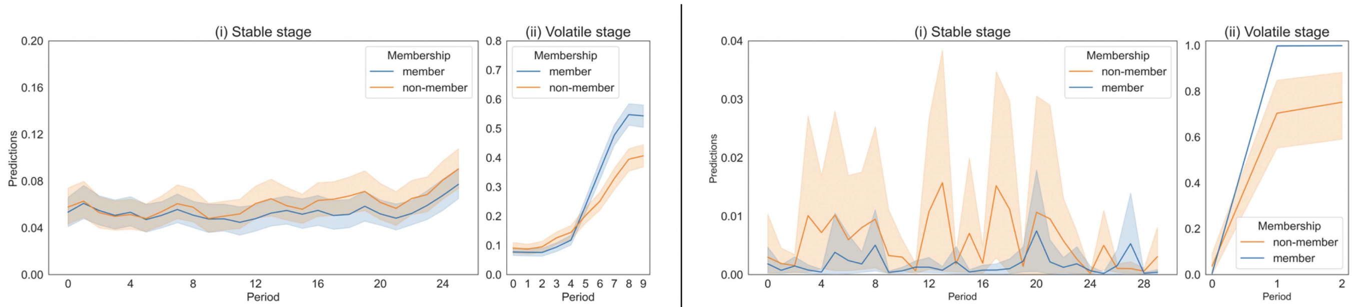
where f_θ is a time-series prediction model that learns temporal patterns from historical observations to generate a prediction $\hat{\mathrm{y}}$, with T denoting the length of the observation sequence.

The primary goal of time-series predictive analytics is to capture intrinsic temporal dependencies within historical temporal sequences. With the advancement of deep learning techniques, deep learning-based time-series prediction models have gained prominence [20], enabling the extraction of more complex and flexible temporal patterns to improve predictive accuracy. Inspired by patch-based architectures in CV [33] and NLP [34] domains, recent researches explore patch-level temporal dependencies within time-series data to enhance local temporal pattern recognition [22], [23], [35]. In this study, we investigate the membership leakage risks of time-series prediction models, where an adversary can exploit a record to generate a sequence of predicted outputs for conducting MIAs, a setting that remains underexplored in prior research. Notably, temporal correlations within predicted output sequences may amplify membership leakage risks in certain models, potentially exceeding the levels estimated by existing MIA methods.

III. Problem Formulation

Target Model. In this work, we consider a length-variant time-series prediction model as the target that an adversary aims to attack. Let the training dataset be $\mathcal{D}^{\mathcal{T}} = \{r\}$, where each record $r = \{(\mathrm{x}_t, \mathrm{y}_t)\}_{t=1}^{T}$ contains a sequence of feature-value pairs for an object over T time steps. To facilitate the learning of length-variant temporal dependencies and enrich the training data, each record is decomposed into multiple training samples of the form $\{\mathrm{x}_1, \ldots, \mathrm{x}_t, \mathrm{y}_{t+1} | t = 1, 2, \ldots\}$. These samples are then used to train the target time-series prediction model, denoted as:

$$f_\theta^{\mathcal{T}} : \{\mathrm{x}_1, \ldots, \mathrm{x}_t\} \rightarrow \hat{\mathrm{y}}_{t+1}, \quad \forall t \in \{1, 2, \ldots\}$$



(a) MIMIC **(b) Fraud**

Fig. 2. Visualization analysis on the MIMIC and Fraud datasets.

TABLE I
KEY NOTATIONS USED IN THE FRAMEWORK

Notation	Description
$f_\theta^{\mathcal{T}}$	The target model
$f_{\theta,i}^{S}$	The i-th shadow model
$\mathcal{A}$	The adversary's attack model
$\mathcal{C}$	The set of patch-based classifiers, $\mathcal{C} = \{\mathcal{C}_1, \ldots, \mathcal{C}_{N_\mathcal{C}}\}$
$\mathcal{D}^{\mathcal{T}}$	The training dataset of the target model $f_\theta^{\mathcal{T}}$
$\mathcal{D}^{Aux}$	The adversary's auxiliary dataset sharing the same distributional properties with $\mathcal{D}^{\mathcal{T}}$
$\mathcal{D}_i^{S}$	The corresponding shadow dataset of shadow model $f_{\theta,i}^{S}$
$\mathcal{D}_i^{S,M}$	The member subset of $\mathcal{D}_i^{S}$
$\mathcal{D}_i^{S,NM}$	The non-member subset of $\mathcal{D}_i^{S}$
$\mathcal{D}^{\mathcal{A}}$	The attack dataset used to train the attack model $\mathcal{A}$
r	A sequential record with the input-output pair in T over T time steps, $r = (\mathrm{X}, \mathrm{Y}) = \{(\mathrm{x}_t, \mathrm{y}_t)\}_{t=1}^{T}$
$\hat{\mathcal{Y}}$	The generated prediction sequence, $\hat{\mathcal{Y}} = \{\hat{\mathrm{y}}_i\}_{i=1}^{N}$
$\mathcal{Y}$	The corresponding actual values of $\hat{\mathcal{Y}}$, $\mathcal{Y} = \{\mathrm{y}_i\}_{i=1}^{N}$
$\bar{\mathcal{Y}}$	The differences between $\mathcal{Y}$ and $\hat{\mathcal{Y}}$, $\bar{\mathcal{Y}} = \{\bar{\mathrm{y}}_i\}_{i=1}^{N}$ where $\bar{\mathrm{y}}_i = \mathrm{y}_i - \hat{\mathrm{y}}_i, \forall i \in 1, \ldots, N$

This training strategy is widely adopted in real-world scenarios, including ICU decompensation prediction and sequential recommendation systems. As we will demonstrate later, time-series models trained in this manner are particularly susceptible to MIAs.

Adversary's Goal. Given a sequential record $r = (X, Y) = \{(\mathrm{x}_t, \mathrm{y}_t)\}_{t=1}^{T}$, the adversary aims to determine whether r was included in the target dataset $\mathcal{D}^{\mathcal{T}}$, which was used to train the target model $f_\theta^{\mathcal{T}}$.

Adversary's Knowledge. In this study, we focus on MIAs in black-box scenarios where the adversary only has access to the predicted output of the target model [4], [8], [19]. Additionally, we make two assumptions on the adversary's background knowledge of the target model, denoted as $\mathcal{B}$, following previous MIA studies [1], [3], [4], [19]. First, we assume that the adversary has an *auxiliary dataset* $\mathcal{D}^{Aux}$ that follows the same distribution as the target dataset $\mathcal{D}^{\mathcal{T}}$, which is used to train the target model, but contains no overlapping records. Second, we assume that the adversary has knowledge of the target model architecture, denoted by $\mathcal{G}^{\mathcal{T}}$. We also evaluate our method when the adversary uses a different architecture from the target model in the robustness analysis.

While Table I summarizes the important notations used in this study, we define our problem as follows:

Definition 1. Membership Inference Attack Problem against Time-series Prediction Models. Given black-box access to a target time-series prediction model $f_\theta^{\mathcal{T}}$, a sequential record r, and the adversary's background knowledge $\mathcal{B} = \{\mathcal{D}^{Aux}, \mathcal{G}^{\mathcal{T}}\}$, the goal is to learn an attack model $\mathcal{A}$ that predicts whether r was used to train the target model $f_\theta^{\mathcal{T}}$:

$$\mathcal{A} : r, f_\theta^{\mathcal{T}}, \mathcal{B} \rightarrow \{0, 1\} \tag{3}$$

where 1 indicates that r was included in the target dataset $\mathcal{D}^{\mathcal{T}}$ (i.e., r is a member), and 0 otherwise.

IV. EMPIRICAL ANALYSIS

In this section, we conduct a visualization analysis to explore universally present temporal discrepancies in sequential outputs between members and non-members for more effective MIAs. We use the MIMIC and Fraud datasets as examples (see Section VI-A for details). For each dataset, two non-overlapping subsets of equal size are sampled to train two target models, with one subset used for training and the other for validation. We then collect sequential outputs from both models for all records. To generate the visualizations, prediction sequence segments with identical labels are extracted from all records and grouped into members and non-members according to the membership status of the record. Fig. 2 displays the averaged predicted probability scores for the positive class, where blue and orange curves denote members and non-members, respectively. Shaded regions indicate 95% confidence intervals, capturing variability across records.

For the MIMIC dataset, we select a stable segment with all-zero labels (Fig. 2(a.i)) and a volatile segment whose labels change from zero to one at period 5 (Fig. 2(a.ii)). Distinct temporal discrepancies between members and non-members emerge in both stages. In the volatile stage, as labels shift from 0 to 1, member predictions are closer to 0 before the transition and quickly adjust to 1. In contrast, non-member predictions adapt more slowly, exhibiting a delayed and gradual rise toward 1. In the stable stage, member predictions exhibit smaller fluctuations across adjacent periods, indicating greater stability than non-members. Similarly, for the Fraud dataset, we select a stable segment with all-zero labels (Fig. 2(b.i)) and a volatile segment with a label sequence [0,1,1] (Fig. 2(b.ii)). The discrepancies observed above also exist: members adjust more rapidly to label

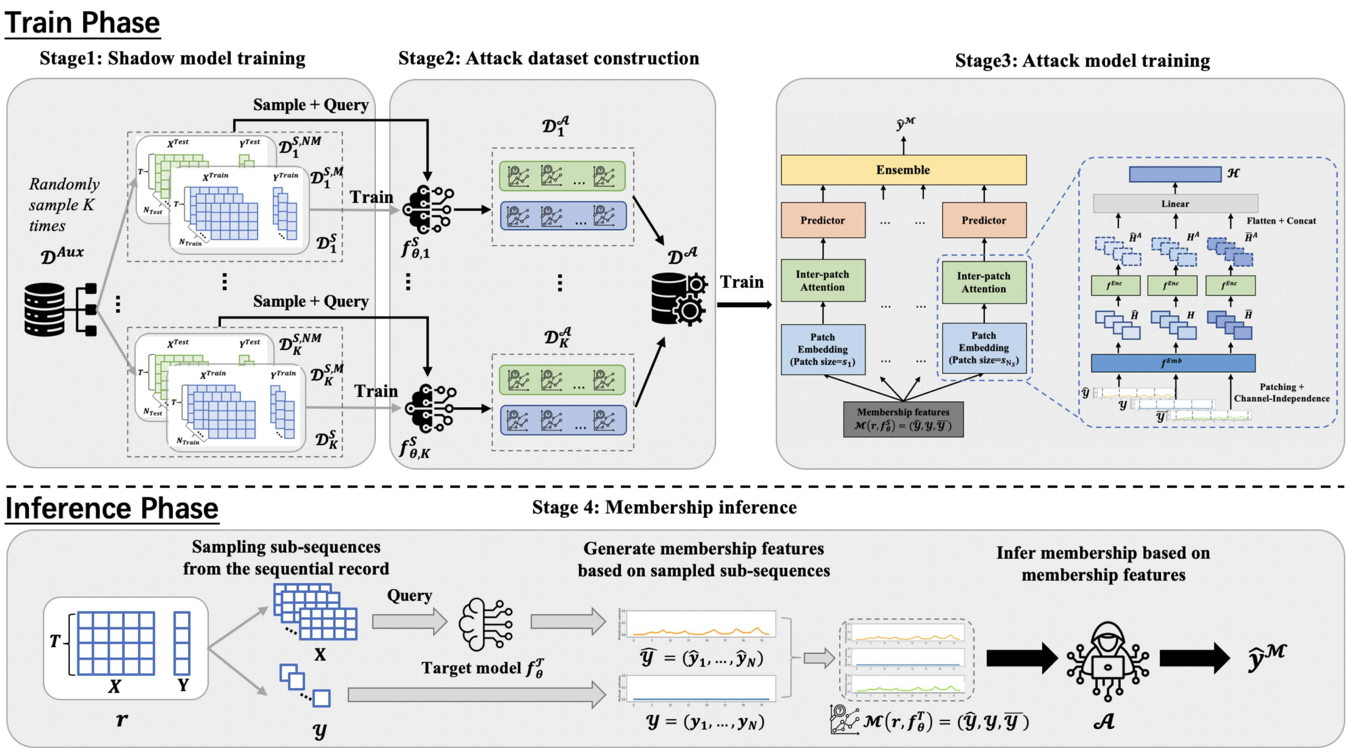


Fig. 3. Overview of TSP-MIA.

changes in volatile stages, and they exhibit smaller fluctuations than non-members in stable stages.

The success of MIAs is often attributed to output discrepancies between members and non-members, resulting from model memorization in non-time-series contexts. We highlight that the temporal discrepancies we identify extend this concept specifically to time-series models, where sequential predictions for member records closely follow the actual values due to the model's memorization of training data (i.e., members). Consequently, predictions for member records exhibit greater stability when actual values remain constant. Conversely, when actual values change, predictions for member records adjust more swiftly to these transitions, reflecting the model's memorization of similar temporal patterns encountered during training. In contrast, predictions for non-member records often lag behind or deviate from the actual changes.

The above analysis reveals that, beyond global temporal patterns which are defined as long-range dependencies across the entire prediction sequence (e.g., trends and seasonality, as discussed in [17]), members and non-members also exhibit distinct local temporal patterns. These patterns reflect short-term behaviors of predicted outputs over consecutive steps, such as the smaller fluctuations and faster adjustments observed in member records. These findings thus motivate the development of a tailored framework that jointly models global and local temporal discrepancies, enabling more effective MIAs and more accurate evaluation of membership leakage risks.

V. Attack Methodology

In this section, we introduce *TSP-MIA*, a novel **M**embership **I**nference **A**ttack framework against **T**ime-**S**eries **P**rediction models. The whole pipeline is shown in Fig. 3, which consists of four main steps: (1) shadow model training, (2) attack dataset construction, (3) attack model training, and (4) membership inference.

A. Shadow Model Training

Following previous classifier-based MIAs [1], [19], [30], TSP-MIA first trains multiple shadow models $\{f^S_{\theta,1}, \ldots, f^S_{\theta,K}\}$ to mimic the behavior of the target model. These shadow models are then utilized to generate training data for the attack model, incorporating both member and non-member samples in the subsequent stage of the process. Specifically, given the adversary's background knowledge about the target model, represented as $\mathcal{B} = \{\mathcal{D}^{Aux}, \mathcal{G}^{\mathcal{T}}\}$, a shadow dataset $\mathcal{D}^S_i$ is first randomly sampled from the auxiliary dataset $\mathcal{D}^{Aux}$ and partitioned into a member subset $\mathcal{D}^{S,M}_i$ for training and a non-member subset $\mathcal{D}^{S,NM}_i$ for validation. A shadow model $f^S_{\theta,i}$, adopting the same architecture $\mathcal{G}^{\mathcal{T}}$ as the target model, is then trained on the member subset $\mathcal{D}^{S,M}_i$ and evaluated on the non-member subset $\mathcal{D}^{S,NM}_i$. The adversary collects the model outputs for both member and non-member records, which are subsequently aggregated across all shadow models to form the training data for the attack model.

B. Attack Dataset Construction

Given the trained shadow models and corresponding shadow datasets, we aim to construct an attack dataset $\mathcal{D}^{\mathcal{A}}$ for training the attack model. To this end, for each sequential record in a shadow dataset, we first sample sub-sequences from the raw feature sequence and obtain their sequential outputs generated from the corresponding shadow model. Then, membership features of the record are extracted using the sequential outputs and corresponding actual values. These features, along with the record's membership status (member or non-member), form an attack sample. Finally, we aggregate all attack samples generated from shadow datasets to construct the attack dataset $\mathcal{D}^{\mathcal{A}}$. In the following, we first elaborate on the membership feature extraction process and then detail the construction of the final attack dataset.

Membership feature extraction. Given a sequential record $r = (X, Y) = \{(\mathrm{x}_t, \mathrm{y}_t)\}_{t=1}^{T}$ in a shadow dataset $\mathcal{D}_i^S$, we aim to extract its membership features based on its sequential outputs as well as corresponding actual values. Specifically, at each time step t, the predicted output $\hat{\mathrm{y}}_t$ is generated by the shadow model $f_{\theta,i}^S$, which takes a sub-sequence $\mathbf{x}^{(t)}$ as input, where the sub-sequence $\mathbf{x}^{(t)}$ is sampled as either $\mathrm{x}_{t-L:t-1}$ or $\mathrm{x}_{1:t-1}$, depending on the temporal prediction task. The corresponding ground-truth is given by y_t. By feeding each sampled sub-sequence into the trained shadow model at each time step, we obtain a sequence of predicted outputs, denoted as $\hat{\mathcal{Y}} = \{\hat{\mathrm{y}}_1, \ldots, \hat{\mathrm{y}}_N\}$ where N represents the total number of sampled sub-sequences. Meanwhile, we collect the corresponding ground-truth value at each step, resulting in a sequence of actual values $\mathcal{Y} = \{\mathrm{y}_1, \ldots, \mathrm{y}_N\}$. Furthermore, to highlight the alignment between predicted and actual values, we compute their point-wise difference at each time step. The resulting sequence of differences is denoted as $\bar{\mathcal{Y}} = \mathcal{Y} - \hat{\mathcal{Y}}$. Finally, the membership features of a record r in shadow dataset $\mathcal{D}_i^S$ are defined as $\mathcal{M}(r, f_{\theta,i}^S)$, as formulated in (4).

$$\mathcal{M}\left(r, f_{\theta,i}^S\right) = \left(\hat{\mathcal{Y}}, \mathcal{Y}, \bar{\mathcal{Y}}\right) \tag{4}$$

where $(\hat{\mathcal{Y}}, \mathcal{Y}, \bar{\mathcal{Y}})$ are the predicted, actual, and difference sequences, respectively. To theoretically justify the design of membership features, let M denote the binary indicator of the membership status of a record. We expect a high mutual information $I(\hat{\mathcal{Y}}, \mathcal{Y}, \bar{\mathcal{Y}}; M)$, indicating that the extracted features are informative about membership. By the chain rule, $I(\hat{\mathcal{Y}}, \mathcal{Y}, \bar{\mathcal{Y}}; M) = I(\hat{\mathcal{Y}}, \mathcal{Y}; M) + I(\bar{\mathcal{Y}}; M \mid \hat{\mathcal{Y}}, \mathcal{Y})$. The first term, $I(\hat{\mathcal{Y}}, \mathcal{Y}; M)$, captures the contribution of prediction and actual sequences to global temporal discrepancies. Prior work [17] shows that global temporal patterns (e.g., trend and seasonality) within these sequences differ between members and non-members. Consequently, including $(\hat{\mathcal{Y}}, \mathcal{Y})$ enables capturing such global differences. The second term, $I(\bar{\mathcal{Y}}; M \mid \hat{\mathcal{Y}}, \mathcal{Y})$ quantifies the incremental contribution of the difference sequence $\bar{\mathcal{Y}}$, which explicitly isolates deviations between $\hat{\mathcal{Y}}$ and $\mathcal{Y}$. These deviations expose local temporal discrepancies, such as smaller fluctuations and faster adjustments in member records, that are less visible when directly given $\hat{\mathcal{Y}}$ and $\mathcal{Y}$. Taken together, the global differences captured by $(\hat{\mathcal{Y}}, \mathcal{Y})$ and the local deviations emphasized by $\bar{\mathcal{Y}}$ provide complementary information, enhancing the exploitation of temporal discrepancies between members and non-members.

Attack dataset construction. For each shadow model $f_{\theta,i}^S$ and the shadow dataset $\mathcal{D}_i^S$, we can obtain an attack subset $\mathcal{D}_i^{\mathcal{A}}$ which consists of membership feature-status pairs:

$$\mathcal{D}_i^{\mathcal{A}} = \left\{\left(\mathcal{M}\left(r, f_{\theta,i}^S\right), 1\right) | r \in \mathcal{D}_i^{S,M}\right\} \cup \left\{\left(\mathcal{M}\left(r, f_{\theta,i}^S\right), 0\right) | r \in \mathcal{D}_i^{S,NM}\right\} \tag{5}$$

Finally, we aggregate all subsets $\{\mathcal{D}_1^{\mathcal{A}}, \ldots \mathcal{D}_K^{\mathcal{A}}\}$ to get the final attack dataset $\mathcal{D}^{\mathcal{A}}$ for attack model training.

$$\mathcal{D}^{\mathcal{A}} = \mathcal{D}_1^{\mathcal{A}} \cup \cdots \cup \mathcal{D}_K^{\mathcal{A}} \tag{6}$$

C. Attack Model Training

Given the attack dataset $\mathcal{D}^{\mathcal{A}}$, we aim to build a binary attack model $\mathcal{A}$ to conduct MIAs. As suggested by the empirical analysis, beyond differences in global temporal patterns, the attack model should also capture the observed local distinctions between members and non-members to mitigate the underestimation of membership leakage risks. To this end, we introduce a tailored patch-based attack model designed to capture both local and global temporal patterns in the sequential outputs to enhance MIA. Specifically, we first train multiple patch-based membership classifiers, each associated with a specific patch size. Then, we ensemble predictions from multiple patch-based classifiers to enhance MIA performance. We illustrate the details of patch-based membership classifiers and the ensemble process in the following sections.

1) Patch-Based Membership Classifier: To leverage both local and global temporal patterns for MIAs, we propose a patch-based membership classifier comprising: (1) a Patch Embedding (PE) module for local feature encoding, and (2) an Inter-Patch Attention (IA) module for modeling temporal dependencies across patches, as well as (3) a binary classification head. Specifically, given the membership features of a record $\mathcal{M}(r, f_\theta^S) = (\hat{\mathcal{Y}}, \mathcal{Y}, \bar{\mathcal{Y}})$, we follow the channel independence settings in [23], [35] to mitigate overfitting issues. In this case, each sequence in the membership features is processed by the PE and IA modules independently. Their representations are finally concatenated as the final embedding, which is then fed into the binary classification head to generate the predicted membership status. In the following, we take the prediction sequence $\hat{\mathcal{Y}}$ as an example to illustrate how the PE and IA modules together capture local and global temporal patterns.

Patch Embedding (PE) module. General temporal models like RNN and LSTM may overlook critical local patterns that differentiate members from non-members, leading to suboptimal MIA performance. Inspired by recent patch-based techniques [22], [23], [35], we generate patch embeddings of membership features to capture local temporal patterns within each patch, thereby enhancing MIA effectiveness. Specifically, given a patch size s, each sequence in the membership features is divided into P patches. The local pattern within each patch is encoded via a linear embedding layer, denoted as f^{Emb}. Taking the prediction sequence $\hat{\mathcal{Y}}$ as an example, the patching process produces a sequence of patches $\{\hat{\mathcal{Y}}^{(1)}, \ldots, \hat{\mathcal{Y}}^{(P)}\}$, where each patch $\hat{\mathcal{Y}}^{(i)}$ contains the predictions within s consecutive time steps. Each patch is then passed through a linear embedding layer f^{Emb} to generate the corresponding patch embeddings:

$\hat{h}_i = f^{\text{Emb}}(\hat{\mathcal{Y}}^{(i)}), \quad \forall i \in \{1, \ldots, P\}$. The resulting sequence of patch embeddings for $\hat{\mathcal{Y}}$ is denoted as $\hat{H} = \{\hat{h}_1, \ldots, \hat{h}_P\}$. Similarly, we obtain the patch embedding sequences H and $\bar{H}$ by applying the same embedding layer f^{Emb} to $\mathcal{Y}$ and $\bar{\mathcal{Y}}$, respectively.

Inter-patch Attention (IA) module. Global temporal pattern discrepancies have proven effective in MIAs. However, Koren et al. [17] only capture trend and seasonality patterns via hand-crafted features, thus failing to capture more complex global temporal patterns that differentiate members from non-members. To address this limitation, the IA module applies an encoder f^{Enc} based on the multi-head self-attention mechanism to the patch embedding sequence $\hat{H}$. For each head $j \in \{1, \ldots, n_h\}$, query, key, and value matrices are obtained through linear projections: $\hat{Q}^{(j)} = \hat{H}W_Q^{(j)}, \hat{K}^{(j)} = \hat{H}W_K^{(j)}, \hat{V}^{(j)} = \hat{H}W_V^{(j)}$. The attention for head j is then computed as: $\text{Attn}^{(j)} = \text{Softmax}(\hat{Q}^{(j)}\hat{K}^{(j)T})\hat{V}^{(j)}$. Outputs from all heads are concatenated to form the adjusted representation: $\hat{H}^A = \text{Concat}(\text{Attn}^{(1)} \mid \ldots \mid \text{Attn}^{(n_h)})$. Similarly, the adjusted representations of the actual sequence $\mathcal{Y}$ and the difference sequence $\bar{\mathcal{Y}}$ can be generated in the same manner, i.e., $H^A = f^{\text{Enc}}(H), \bar{H}^A = f^{\text{Enc}}(\bar{H})$. While the PE module preserves critical local patterns via patch embedding generation, the IA module complements it by capturing global dependencies across patches. The generated representations thus provide a holistic view of both local and global patterns essential for effective membership inference.

Learning objective. Given the adjusted representations $\tilde{H}^A$, H^A, and $\bar{H}^A$, the final representation of a record $\mathcal{H}$ is formed by flattening and concatenating them, followed by a linear transformation. This representation is then passed to a binary classification head, producing the membership confidence vector $\hat{y}^{\mathcal{M}} = (\hat{y}_0^{\mathcal{M}}, \hat{y}_1^{\mathcal{M}})$, where $\hat{y}_0^{\mathcal{M}}$ and $\hat{y}_1^{\mathcal{M}}$ denote the probabilities of non-membership and membership, respectively. The classifier is trained using binary cross-entropy loss: $L = -\sum_{i=1}^{|\mathcal{D}^{\mathcal{A}}|}(y_i^{\mathcal{M}} \log \hat{y}_{i,1}^{\mathcal{M}} + (1 - y_i^{\mathcal{M}}) \log \hat{y}_{i,0}^{\mathcal{M}})$. where $y_i^{\mathcal{M}}$ is the real membership status of the i^{th} record, and $|\mathcal{D}^{\mathcal{A}}|$ is the size of the attack dataset $\mathcal{D}^{\mathcal{A}}$.

2) Multi-Scale Ensemble: Local temporal patterns indicative of membership status may emerge at different time resolutions. To exploit this, we propose a multi-scale ensemble strategy that integrates temporal patterns captured at multiple granularities to enhance MIA performance. As illustrated above, each patch-based membership classifier is associated with a specific patch size, enabling it to focus on temporal patterns at a specific resolution. To capture a broader range of temporal patterns, we train a set of $N_{\mathcal{C}}$ patch-based classifiers, denoted as $\mathcal{C} = \{\mathcal{C}_1, \ldots, \mathcal{C}_{N_{\mathcal{C}}}\}$, where each $\mathcal{C}_k$ is associated with a distinct patch size $s_k \in \{s_1, \ldots, s_{N_{\mathcal{C}}}\}$. Given a record r_i, each classifier $\mathcal{C}_k$ produces a membership confidence vector $\hat{y}_i^{\mathcal{M},(k)}$. The ensemble prediction is then computed by averaging these outputs across all classifiers: $\hat{y}_i^{\mathcal{M}} = \frac{1}{N_{\mathcal{C}}}\sum_{k=1}^{N_{\mathcal{C}}} \hat{y}_i^{\mathcal{M},(k)}$. This ensemble strategy allows TSP-MIA to capture and integrate membership-indicative patterns at multiple temporal resolutions, thereby improving its robustness and effectiveness. Note that we only explore the potential of enhancing MIAs by capturing multi-scale local temporal discrepancies and leave more advanced model designs for further studies.

D. Membership Inference

With the trained attack model $\mathcal{A}$, the adversary conducts a membership inference attack (MIA) on a record through the following steps: First, sub-sequences are sampled from the raw feature sequence of the record and fed into the target model to generate sequential outputs. Then, its membership features are extracted based on the sequential outputs and corresponding actual values. Finally, membership features are input into the trained attack model for membership inference.

E. Complexity and Scalability Analysis

Given the length of membership features N and a patch embedding dimension d, the computational complexity of TSP-MIA is dominated by the IA module's self-attention operation, $\mathcal{O}(N^{2}\tilde{\ }d)$, while the PE module and classification head contribute only $\mathcal{O}(Nd)$. As the number of patch-based classifiers $N_{\mathcal{C}}$ is a hyperparameter, the overall complexity remains $\mathcal{O}(N^{2}\tilde{\ }d)$. The ensemble design scales linearly with $N_{\mathcal{C}}$, as classifiers are trained independently and only their predictions are combined. This allows training and inference to be fully parallelized across devices, eliminating sequential bottlenecks and enabling a flexible trade-off between computational cost and the granularity of captured temporal patterns.

VI. Experimental Evaluation

A. Experiment Setup

1) Datasets: We consider two real-world time-series datasets to conduct experiments: a clinical dataset *MIMIC* [36] and a financial dataset *Fraud* [37]. The MIMIC dataset is derived from the public MIMIC-III ICU database [38], following the preprocessing pipeline of [36]. It is used for the decompensation prediction task, which predicts whether a patient will experience decompensation within the next 24 hours using all available EHR measurements up to the current hour. At each hour t, x_t denotes physiological features (e.g., blood pressure, glucose level), and y_t is a binary label indicating decompensation. The Fraud dataset is used for the fraud detection task, which predicts the fraud status of the current transaction using the user's most recent K transactions ($K = 5$ in this study). At step t, x_t represents transaction features (e.g., amount, merchant ID), and y_t indicates whether the transaction is fraudulent.

2) Time-Series Model Architectures: For the target time-series model, we consider four architectures for both datasets, including LSTM [36], Channel-wise LSTM (CW-LSTM) [36], Transformer [39], and DLinear [40]. These architectures represent typical structures for temporal modeling. Additionally, for the MIMIC dataset, we further include StageNet [41], a sequential model tailored for the medical context.

3) Baselines: To illustrate the effectiveness of TSP-MIA, we compare it with three types of state-of-the-art MIA baselines:

single-prediction-based MIAs, multi-prediction-based MIAs, and temporal MIAs.

Single-output-based MIA. The single-output-based baseline, *L-SMIA* [1], extracts the predicted output at the final step and its corresponding ground-truth label as membership features, which are then fed into an MLP-based attack model for membership inference. It serves as the basis for membership leakage risk evaluation.

Multi-output-based MIAs. These baselines, proposed by Shejwalkar et al. [31], leverage discrepancies in multiple generated outputs between members and non-members to infer membership status. Notably, they do not model correlations among different outputs. *T-UMIA* is a metric-based method that infers membership by comparing the average predicted class-specific probability with a preset threshold. *L-UMIA* extracts statistics of the predicted outputs as membership features and then feeds them into an MLP-based attack model for MIA. *STUMIA* also follows a metric-based paradigm but operates at the per-output level. It conducts the MIA using the entropy of predicted probabilities for each output and aggregating these results to make a final prediction.

Temporal MIAs. These approaches capture temporal patterns within sequential outputs to implement MIAs, potentially enhancing attack performance against time-series models. *TS-MIA* [17] derives two temporal membership features, i.e., trend and seasonality, and then feeds them into an MLP-based attack model for membership inference. *LSTM-MIA* follows the same framework as TSP-MIA but employs an LSTM-based attack model.

Additionally, we define a variant of TSP-MIA, referred to as *TSP-MIA-S*, which employs only a single patch-based membership classifier and does not utilize the multi-scale ensemble strategy.

4) Metrics: Following state-of-the-art works [3], [4], [5], we first employ the widely adopted average-case metric, balanced accuracy, which quantifies the likelihood of a membership inference attack (MIA) correctly identifying members and non-members in a balanced dataset. Additionally, we incorporate *TPR @ low FPR* and *Full Log-scale ROC*, two metrics recommended by Carlini et al. [3]. *TPR @ low FPR* measures the true-positive rate at a fixed low false-positive rate (e.g., 1% FPR as in Koren et al. [17]), offering a concise evaluation of attack effectiveness under stringent conditions. *Full Log-scale ROC*, otherwise, provides a comprehensive assessment by plotting ROC curves on a log scale, emphasizing TPRs in low FPR regions for a detailed performance analysis. To ensure robustness, we conduct 10 experimental runs and report the average values of these metrics.

5) Implementation Details: Dataset Partition. To implement MIAs on both the MIMIC and Fraud datasets, we train 10 shadow models ($K = 10$), each using a separate shadow dataset $\mathcal{D}_i^S$. For the Fraud dataset, each shadow dataset contains 512 samples, evenly split into 256 member and 256 non-member samples. For the MIMIC dataset, each shadow dataset consists of 1024 samples, with 512 members and 512 non-members. The number of sampled sub-sequences per sample (N) is 200. To construct the attack dataset, we create balanced attack datasets with 5000 samples for MIMIC and 5120 samples for Fraud. Additionally, the target model's training and test datasets each contain 256 samples for Fraud and 512 samples for MIMIC, ensuring a balanced validation dataset for attack models.

Target model preparation. We construct target models using the default parameter settings for StageNet and DLinear across both datasets. For LSTM, the hidden dimension is set to 384 for MIMIC and 100 for Fraud. For CW-LSTM and Transformer, the hidden size is set to 32 for both datasets. Training is conducted for 15 to 100 epochs on the MIMIC dataset and 50 to 250 epochs on the Fraud dataset, depending on dataset complexity and model architecture.

Attack model training. We reimplemented baselines according to the original references [1], [17], [31] and fine-tune hyper-parameters for performance optimization. Specifically, the MLP-based attack models employed in L-SMIA, L-UMIA, and TS-MIA share the same architecture with two MLP layers of sizes 32 and 16, while LSTM-MIA consists of two LSTM layers with a hidden size of 128. For TSP-MIA-S and TSP-MIA, we fix the number of attention heads n_h as 4 and then fine-tune the patch size s and the patch embedding size d of the patch-based classifier via grid search, where s and d are selected from $\{2, 3, 4, 6, 8, 12\}$ and $\{32, 64, 128\}$, respectively. Since TSP-MIA leverages multiple patch-based membership classifiers, we also fine-tune the number of classifiers $N_{\mathcal{C}}$ from 2 to 5. During attack model training, we adopt the Adam optimizer [42] with the learning rate of 1e-3 and weight decay set to 5e-4, and a total of 50 training epochs. For our proposed attack models, we adjust the learning rate to 5e-4 and extend training to 100 epochs due to their increased complexity. Additionally, we apply early stopping with a patience of 10 epochs (monitoring validation loss) to prevent overfitting across all attack models. For all attack models, we adopted a batch size of 64. All experiments are conducted on a computing cluster equipped with four NVIDIA Tesla V100S GPUs (32 GB each) and two NVIDIA Tesla V100 GPUs (16 GB each).

B. Research Questions

In the following sections, we aim to investigate the following research questions: (*RQ1.*) To what extent do the uncovered temporal discrepancies in sequential outputs amplify membership leakage risks in time-series prediction models? (*RQ2.*) How do different components of TSP-MIA influence its effectiveness for membership leakage risk evaluation? (*RQ3.*) How does the adversary's background knowledge of the target model influence the effectiveness of MIAs?

C. Comparison With Baselines (RQ1)

To address RQ1, we evaluate the performance of TSP-MIA against various state-of-the-art MIA baselines on real-world datasets and target models with varying architectures. Tables II and III present a comparative analysis in terms of balanced accuracy and TPR@1% FPR on the MIMIC and Fraud datasets, respectively. Additionally, Figs. 4 and 5 illustrate the full log-scale ROC curves for different datasets, providing a comprehensive visualization of attack performance. In contrast, our findings

TABLE II
BASELINE COMPARISONS ON THE MIMIC DATASET

Attack method	Balanced Accuracy(%)					TPR@1%FPR(%)				
	LSTM	CW-LSTM	StageNet	Transformer	DLinear	LSTM	CW-LSTM	StageNet	Transformer	DLinear
L-SMIA	50.84	52.65	68.60	56.08	58.33	0.59	0.45	0.00	2.50	1.95
T-UMIA	50.59	50.00	50.00	60.06	49.95	0.00	0.00	0.00	0.00	0.00
L-UMIA	51,62	52.98	52.10	52.11	52.14	2.03	0.41	0.70	1.84	0.59
STUMIA	50.49	50.10	50.59	61.13	50.20	0.00	0.00	0.00	0.00	0.00
TS-MIA	50.02	55.06	70.28	58.44	52.94	0.00	0.00	0.00	0.00	0.00
LSTM-MIA	56.97	53.99	54.25	54.71	53.48	1.25	1.19	1.39	1.41	1.66
TSP-MIA-S	93.79	73.24	69.96	87.71	**90.94**	58.57	6.02	**11.26**	51.21	**65.76**
TSP-MIA	**95.47**	**74.10**	**70.61**	**88.17**	89.56	**81.33**	**7.38**	10.35	**56.21**	64.84

TABLE III
BASELINE COMPARISON ON THE FRAUD DATASET

Attack method	Balanced Accuracy(%)				TPR@1%FPR(%)			
	LSTM	CW-LSTM	Transformer	DLinear	LSTM	CW-LSTM	Transformer	DLinear
L-SMIA	50.12	50.20	50.18	50.12	0.66	0.27	1.33	0.59
T-UMIA	52.54	55.08	51.37	52.93	0.00	0.00	0.00	0.00
L-UMIA	51.05	52.05	51.66	51.72	2.27	2.89	2.03	0.43
STUMIA	50.00	50.00	50.00	50.00	0.00	0.00	0.00	0.00
TS-MIA	75.00	75.00	58.52	58.40	3.48	1.76	1.29	0.74
LSTM-MIA	63.87	65.20	51.09	49.96	3.95	9.30	1.87	1.48
TSP-MIA-S	78.05	74.16	**62.21**	56.31	12.62	10.74	2.77	2.77
TSP-MIA	**78.67**	**75.47**	61.76	**62.93**	**16.17**	**13.52**	**3.16**	**3.59**

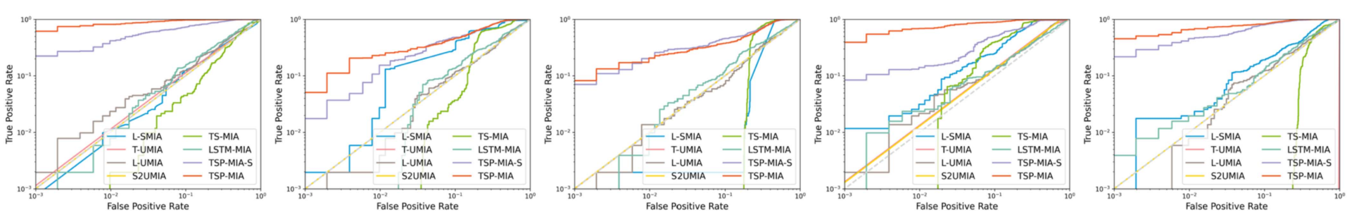


Fig. 4. ROC curves on the MIMIC dataset (from left to right: LSTM, CW-LSTM, StageNet, Transformer, DLinear).

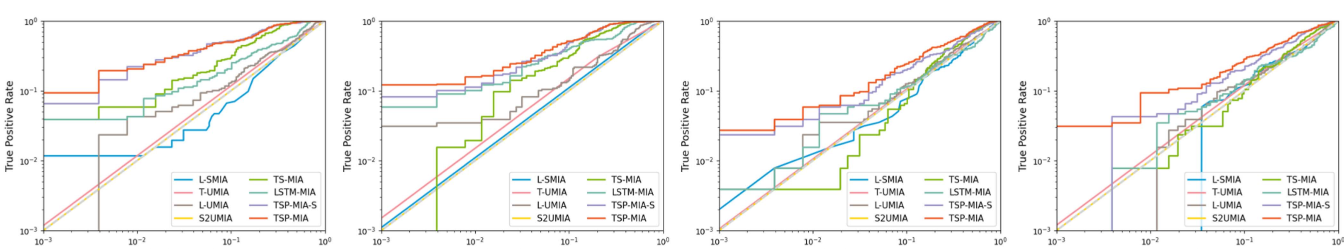


Fig. 5. ROC curves on the Fraud dataset (from left to right: LSTM, CW-LSTM, Transformer, DLinear).

reveal that non-time-series baselines (i.e., single-output-based and multi-output-based MIAs) achieve only slightly above 50% accuracy and near-zero TPR@1% FPR in most scenarios, particularly on the MIMIC dataset. As suggested by our previous analysis, temporal discrepancies play a crucial role in MIAs against time-series models, which indicates the limited effectiveness of non-time-series baselines, and thus it is non-trivial to propose tailored methods for precise risk evaluations.

Notably, temporal MIA baselines (i.e., TS-MIA and LSTM-MIA) generally outperform non-time-series baselines. By capturing global temporal discrepancies across entire membership features, these methods greatly improve attack performances on time-series models. However, these baselines fail to fully capture the fine-grained local temporal discrepancies uncovered in the empirical analysis; they still underestimate membership leakage risks in time-series models. Our methods (TSP-MIA and TSP-MIA-S) consistently achieve superior performance, particularly in the low-FPR regime, as demonstrated by the ROC curves and the TPR@1% FPR metric. For instance, on the MIMIC dataset, while baselines achieve a maximum TPR@1% FPR of approximately 2.5% across target models with different architectures, our methods achieve TPR@1% FPRs ranging from

TABLE IV
ABLATION STUDY ON THE MEMBERSHIP FEATURE LENGTH

Dataset	Metrics(%)	Sequence Length(N) 20	50	100	150	200
MIMIC	Balanced Acc.	65.00	68.24	72.44	81.54	**87.71**
	TPR@1%FPR	26.78	27.07	35.45	45.02	**51.21**
Fraud	Balanced Acc.	51.52	59.20	66.39	72.27	**78.05**
	TPR@1%FPR	0.86	3.52	5.35	7.07	**12.62**

5% to 80% , as shown in Table II. Furthermore, as TSP-MIA leverages multi-scale local temporal patterns within membership features, it outperforms its single-scale variant, TSP-MIA-S, in most cases.

The single-output-based baseline, L-SMIA, focuses on the differences between members and non-members in a single output, whereas multi-output-based baselines (i.e., T-UMIA, L-UMIA, and STUMIA) leverage multiple outputs but neglect temporal discrepancies over them. Temporal baselines capture global temporal discrepancies between members and non-members, yet overlook local temporal discrepancies identified in our empirical analysis. By addressing these limitations and jointly modeling global temporal patterns and multi-scale local temporal patterns, TSP-MIA achieves superior performance across all evaluated settings.

D. Ablation Study (RQ2)

In this section, we examine the impacts of different components of TSP-MIA on the attack performance.

Membership feature length The length of membership features directly determines the extent of temporal discrepancies that can be leveraged for MIAs. Intuitively, longer membership features capture more pronounced temporal discrepancies, thereby enhancing attack effectiveness. We conduct MIAs on the MIMIC dataset using Transformer-based target models and on the Fraud dataset using LSTM-based target models, varying the membership feature length N to examine its effect on attack performance. As shown in Table IV, longer membership features generally lead to better attack performance in terms of both balanced accuracy and TPR@1% FPR, likely due to the greater discrepancies between members and non-members. Notably, on the MIMIC dataset, although the attack performance of TSP-MIA declines as N is reduced from 200 to 20 (i.e., balanced accuracy and TPR@1% FPR dropping to 65.00% and 26.78%), it still outperforms all baselines, whose highest balanced accuracy and TPR@1% FPR are 61.13% (STUMIA) and 2.50% (L-SMIA), respectively.

Membership feature composition In TSP-MIA, the sequence composition of membership features determines how temporal discrepancies between members and non-members are displayed. By default, membership features consist of the predicted sequence ($\hat{\mathcal{Y}}$), the actual sequence ($\mathcal{Y}$), and the difference sequence ($\bar{\mathcal{Y}}$). To evaluate the impacts of different sequence compositions, we conduct MIAs against a CW-LSTM-based target model using the MIMIC dataset, varying the combinations of these sequences as membership features. As shown in Table V , the combination of predicted ($\hat{\mathcal{Y}}$) and actual ($\mathcal{Y}$) sequences

TABLE V
ABLATION STUDY ON THE MEMBERSHIP FEATURE COMPOSITION

Compositions $\hat{\mathcal{Y}}$	$\mathcal{Y}$	$\bar{\mathcal{Y}}$	Balanced Acc. (%)	TPR@1%FPR (%)
×			**74.62**	4.18
		×	69.84	2.71
	×	×	67.28	1.91
×		×	74.10	1.46
×	×		73.56	4.49
×	×	×	73.24	**6.02**

TABLE VI
ATTACK PERFORMANCE ON DIFFERENT TARGET DATASET DISTRIBUTIONS

Target model	AUROC(%) *Low*	*High*	TPR@1%FPR(%) *Low*	*High*
LSTM	87.12	89.63	**16.17**	4.84
CW-LSTM	83.11	87.28	13.52	**14.49**
Transformer	87.55	87.75	**3.16**	1.29
DLinear	87.46	84.14	**3.59**	0.98

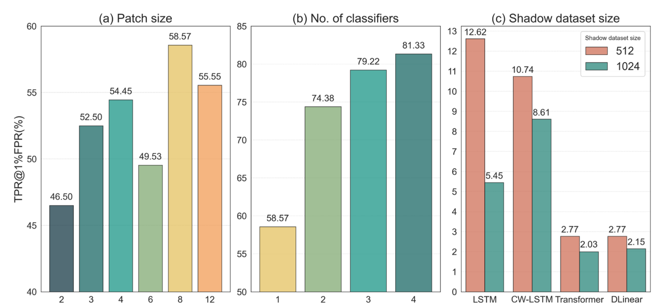


Fig. 6. Ablation study on patch size selection and shadow dataset size.

already provides informative membership features, achieving high balanced accuracy and TPR@1% FPR. Furthermore, incorporating the difference sequence ($\bar{\mathcal{Y}}$) significantly improves TPR@1% FPR, with only a minor reduction in balanced accuracy. These findings highlight the importance of integrating all three sequences as membership features, as they collectively enhance the attack model's ability to distinguish members from non-members.

Patch size selection. This experiment examines how patch size selection affects the attack performance of TSP-MIA on the MIMIC dataset using LSTM-based target models. We first evaluate the single-classifier variant (TSP-MIA-S) under different patch sizes, and then analyze the effects of varying patch size sets $\{s_k\}_{k=1}^{N_{\mathcal{C}}}$ and the number of classifiers $N_{\mathcal{C}}$, with patch embedding sizes fine-tuned accordingly. Fig. 6(a) shows that TSP-MIA-S achieves the best performance when the patch size is 8, indicating that both overly short and overly long patch sizes degrade performance. Fig. 6(b) indicates that attack performance improves with larger $N_{\mathcal{C}}$, though the marginal gains gradually diminish. Notably, optimal patch size sets for larger $N_{\mathcal{C}}$ consistently include those identified for smaller ones (e.g., $\{6, 8\}$ for $N_{\mathcal{C}} = 2$ is contained in $\{6, 8, 12\}$ for $N_{\mathcal{C}} = 3$). Based on these results, we suggest an efficient patch size selection strategy: first train a single classifier with the patch size tuned for optimal performance, and then incrementally add classifiers

TABLE VII
PERFORMANCE COMPARISON AMONG DIFFERENT COMBINATIONS OF SHADOW AND TARGET MODEL ARCHITECTURES *(MIMIC)*

Shadow Architecture	Balanced Acc. (%) Target architecture LSTM	CW-LSTM	StageNet	Transformer	DLinear	TPR@1%FPR(%) Target architecture LSTM	CW-LSTM	StageNet	Transformer	DLinear
LSTM	**93.79**	**74.24**	52.03	70.34	70.05	**58.27**	4.41	1.07	2.09	5.06
CW-LSTM	63.77	<u>73.24</u>	49.34	74.79	73.84	<u>10.57</u>	**6.02**	0.55	17.71	10.88
StageNet	50.05	52.64	**69.96**	51.02	42.58	0.45	0.10	**11.66**	5.12	0.00
Transformer	56.03	51.80	57.71	**87.75**	75.65	0.25	0.82	2.42	**51.25**	17.27
DLinear	51.67	48.29	49.28	53.47	**90.94**	0.33	2.64	1.35	8.89	**65.76**
MIX	<u>76.26</u>	65.23	<u>61.43</u>	<u>77.55</u>	<u>86.05</u>	6.39	<u>4.47</u>	<u>5.94</u>	<u>28.75</u>	<u>39.73</u>

TABLE VIII
PERFORMANCE COMPARISON AMONG DIFFERENT COMBINATIONS OF SHADOW AND TARGET MODEL ARCHITECTURES *(FRAUD)*

Shadow Architecture	Balanced Acc. (%) Target architecture LSTM	CW-LSTM	Transformer	DLinear	TPR@1%FPR(%) Target architecture LSTM	CW-LSTM	Transformer	DLinear
LSTM	**78.67**	<u>71.72</u>	58.65	59.98	<u>16.17</u>	<u>7.93</u>	3.05	1.87
CW-LSTM	74.59	**75.47**	53.81	57.71	**16.91**	**13.52**	**3.67**	2.11
Transformer	70.82	65.21	**61.76**	<u>60.92</u>	3.71	3.40	<u>3.16</u>	2.11
DLinear	62.36	58.50	55.23	**62.93**	5.16	5.08	1.95	**3.59**
MIX	<u>76.04</u>	69.53	<u>60.02</u>	60.72	9.30	7.03	2.34	<u>2.58</u>

TABLE IX
RESULTS ON DP-SGD WITH DIFFERENT σ

Target model	**Metrics(%)**	**Noise Scale(σ)** 0	0.2	0.5	1
LSTM	(↑)AUROC	**76.55**	43.48	46.89	49.93
	(↓)TPR@1%FPR	58.57	**0.59**	0.63	0.70
CW-LSTM	(↑)AUROC	**84.51**	57.06	51.63	48.96
	TPR@1%FPR	6.02	**0.68**	0.96	0.86
StageNet	(↑)AUROC	**83.99**	62.18	63.79	63.73
	(↓)TPR@1%FPR	11.26	3.83	**1.05**	1.07
Transformer	(↑)AUROC	**83.59**	55.16	48.03	45.78
	(↓)TPR@1%FPR	51.25	**14.14**	14.67	14.18
DLinear	(↑)AUROC	**96.12**	63.60	61.98	58.48
	(↓)TPR@1%FPR	65.76	**0.88**	1.13	1.43

with new patch sizes until performance plateaus. This strategy reduces the parameter search complexity from $\mathcal{O}((N_p N_d)^{N_C})$ to $\mathcal{O}(N_p N_d N_C)$, where N_p and N_d are the numbers of candidate patch sizes and embedding sizes, thereby improving efficiency.

Shadow dataset size. Prior studies [1], [43] have shown that larger training sets generally reduce overfitting and thereby lower membership leakage risks. To examine this effect, we conduct MIAs using different shadow dataset sizes on the Fraud dataset. As shown in Fig. 6(c), attack performance consistently declines as dataset size increases, aligning with the expectation that larger datasets reduce membership leakage risks. However, the attacks remain effective, underscoring the robustness of TSP-MIA.

E. Robustness Analysis (RQ3)

In this section, we examine the impact of the adversary's knowledge of the target model on attack performance.

Target dataset distribution. This analysis examines how distribution shifts between shadow and target datasets affect the attack performance of TSP-MIA. Following Shi et al. [44], we represent distribution shifts using label shifts, i.e., differences in the label proportions of datasets. On the Fraud dataset, we apply the same attack model (trained on shadow models with an average fraud proportion of 56.08%) to target models trained on datasets with varying fraud proportions,, where *Low* and *High* correspond to 57.81% and 100% , respectively.[1] Consistent with previous work [8], [44], Table VI shows that distribution shifts have minimal impact on target model performance (AUROC) but noticeably affect attack performance (TPR@1% FPR), which decreases as label shifts increase.

Target model architecture. To assess the impact of the knowledge of the target model architecture, we conduct MIAs while varying the architectures of the shadow and target models. As shown in Tables VII and VIII, attack performance is highest when the shadow and target models share the same architecture. Notably, TSP-MIA archives performances comparable to the best ones when the architectures of shadow and target models belong to the same model family (e.g., LSTM and CW-LSTM). This suggests that models with similar architectures may exhibit similar temporal discrepancies between members and non-members. Since adversaries typically lack access to the target architecture, we propose the MIX strategy to mitigate performance degradation: the adversary trains shadow models with diverse architectures and aggregates them to construct the attack model. As shown in the last rows of Tables VII and VIII, *MIX* achieves slightly lower but still competitive performance compared to the ideal case where shadow and target models share the same architecture. This demonstrates the effectiveness of *MIX* and substantial risks even without precise architectural knowledge.

[1] In the Fraud dataset, a record is labeled as fraudulent if any label in its label sequence indicates fraud. The fraud proportion thus corresponds to the proportion of fraudulent records in the dataset

VII. Defenses

We follow prior studies [8], [45] to evaluate TSP-MIA under the widely adopted defense DP-SGD [46], which introduces Gaussian noise to gradients during model training to ensure differential privacy. The noise scale σ is a key hyperparameter controlling the trade-off between privacy and utility, with larger σ values indicating greater noise and higher privacy protection. We assess attack performance on the MIMIC dataset under varying σ values. As shown in Table IX, while DP-SGD substantially reduces membership leakage risks (lower TPR@1% FPR), it also causes severe utility degradation (reduced AUROC) even at small σ. These results highlight the limitations of DP-SGD in balancing privacy preservation and model utility.

VIII. Conclusion

In this paper, we aim to provide a precise evaluation of membership leakage risks in time-series prediction models through MIAs. Our empirical analysis reveals that, other than global temporal discrepancies, overfitting and model memorization also give rise to distinct local temporal discrepancies between members and non-members, amplifying membership leakage risks. To this end, we propose a novel MIA framework, TSP-MIA, that incorporates a specialized patch-based attack model to exploit both global and local temporal discrepancies to further enhance membership leakage risk evaluations. Extensive evaluations demonstrate that TSP-MIA consistently outperforms previous state-of-the-art methods, uncovering substantial membership leakage risks that were previously underestimated. Additionally, we assess the effectiveness of each component within TSP-MIA. Our robustness analysis also reveals that membership leakage risks persist even when adversaries have limited knowledge of the target model. These findings reveal significant privacy risks in time-series predictive analytics, underscoring the importance of incorporating privacy safeguards in model deployment. Regulators should also consider the potential membership leakage risks when formulating policies for time-series machine learning systems. Our study highlights the pressing need for continued research on membership privacy of time-series models to support the development of secure and trustworthy ML applications.

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